

Samuel Jo

SOFTWARE & AI ENGINEER

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SUMMARY

Software & AI engineer who builds and ships **end to end** — LLM orchestration and grounded generation, real-time and 3D frontends, large data pipelines, and the cloud infra and CI/CD in between. **Four products in production**. At home in zero-to-one problems where the product needs to exist yesterday, and AI systems that run unattended and stay correct at scale.

SKILLS

Languages	TypeScript / JavaScript, Python, Java, C#, C++, SQL, R, HTML/CSS
Frontend	React 19, Next.js 15, Vite, Three.js, Tailwind, PWAs, accessibility, Pretext
Real-time / AI	WebRTC (LiveKit), Socket.IO, MediaPipe · LLM orchestration (Groq/Claude/GPT), RAG, structured output & guardrails, ranking/MMR, autonomous agents
Data & DB	SQLite (FTS5, WAL tuning), PostgreSQL, Firebase/Firestore, Supabase; pipelines, schema design
Infra	AWS (EC2/EBS/VPC), Cloudflare, Vercel, Render, VPS + PM2, Docker, GitHub Actions CI/CD, Stripe, Sentry, PostHog

EXPERIENCE

Founder & AI Engineer, **RegWatch** (regwatch.nyc)

Jan 2025 – Present

Architected a full-stack AI property-intelligence SaaS **0→1: 9.6M+ parcels / 42M+ records** across NY, NJ & CT from **86+ government sources**, with sub-100ms search and instant compliance reports.
Shipped a **RAG AI assistant** (grounded, session-persistent, graceful fallback) and automated compliance monitoring across 150+ government pages with AI change detection and structured alerts.
Built single-codebase **multi-brand host routing, 12-locale i18n** (incl. RTL) with a CI translation gate, Stripe tiered billing, and **blue-green zero-downtime deploys** on a VPS (PM2 cluster).
Optimized a **~114 GB SQLite** store (FTS5, WAL, covering indexes); shipped 65+ features via AI-augmented development with human review gates.

General Manager, **Mizumi**

Aug 2019 – Present

Built and shipped **miztips** (miztips.vercel.app) — a tip-distribution **PWA** (React, Firebase) used **every shift by 18+ staff**: role-weighted pooling, PDF exports, real-time sync with offline fallback.
Run a **\$5M+** hospitality operation with 25+ staff; built data-driven inventory forecasting (–15% waste) and 100%-accurate daily revenue reconciliation.

Full Stack Engineer (Contract), **Tegger**

Oct 2020 – Apr 2021

Delivered a Celo-blockchain widget MVP (React, Node.js) with the CTO; earned the **Celo Design Tenet Award**.

Municipal Financial Analyst, **Queens College**

Aug 2018 – Jul 2019

Restructured CAFRs for **10,000+ U.S. municipalities**; automated balance-sheet validation with VBA (–80% processing time).

SELECTED PROJECTS

Crowdtells crowdtells.com · React/TS · Groq · Cloudflare

A news feed driven by prediction markets: **newsworthiness ranking with MMR diversity**, hallucination-proof LLM briefings grounded in market data, and a zero-cost GitHub-native pipeline. 794 automated tests.

Tells TypeScript · Three.js · MediaPipe · LiveKit

Browser 3D Texas Hold'em rendering each player's live webcam face as a **478-point video-textured 3D mesh**; authoritative server poker engine, peer-to-peer media over a LiveKit SFU.

Quant Trading Engine & Autonomous Bot Python · Playwright · Docker

Backtested 80k+ candles / 30k+ Monte Carlo runs; deployed as a cron-driven **autonomous agent** (signal detection, browser-automated execution, Discord alerts) in production on Railway.

EDUCATION & CERTIFICATIONS

B.A. Economics — Queens College, NY · Econometrics, Statistical Methods, Multivariable Calculus

2016 – 2019

CS50x — Harvard · **Data Structures & Algorithms in Java** — University of Helsinki